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Bio. Peichun is a second-year Ph.D. student in Computer Science at the **University of Macau**. Prior to this, he earned both his B.Eng. in Electronic Information Engineering and my M.Sc. in Control Science and Engineering at **Guangdong University of Technology**.

His research focuses on **edge computing** and **distributed learning**. His earlier work explored federated learning, and he is now shifting towards **collaborative large model deployment** and **multi-agentic AI systems** over edge networks.

Education

Aug. 2024 – Present	**Ph.D. Student**, *Advisor: Prof. Yuan Wu*. *University of Macau* Department of Computer and Information Science, Faculty of Science and Technology
Sept. 2018 – June 2021	**Master of Science**, *Advisor: Prof. Rong Yu*. *Guangdong University of Technology* Department of Control Science and Engineering, School of Automation The recipient of *National Scholarship for Graduate Students*
Sept. 2014 – June 2018	**Bachelor of Engineering**, *Advisor: Prof. Rong Yu*. *Guangdong University of Technology* Department of Electronic Information Engineering, School of Automation The recipient of *National Scholarship for Undergraduates*

Selected Publications

- **Peichun Li**, Liping Qian, Haijun Zhang, Zhiguo Shi, and Yuan Wu, "Enabling Multi-Agentic AI in Future Wireless Networks: An End-Edge-Cloud Approach," accepted by *IEEE Network*.
- **Peichun Li**, Liping Qian, Dusit Niyato, Shiwen Mao, and Yuan Wu, "Toward Resource-Efficient Collaboration of Large AI Models in Mobile Edge Networks," *IEEE Network*, vol. 40, no. 3, pp. 51-59, May 2026. (JCR Q1)
- **Peichun Li**, Hanwen Zhang, Yuan Wu, Liping Qian, Rong Yu, Dusit Niyato, and Xuemin Shen, "Filling the Missing: Exploring Generative AI for Enhanced Federated Learning over Heterogeneous Mobile Edge Devices," *IEEE Transactions on Mobile Computing* vol. 23, no. 10, pp. 10001-10015, Oct. 2024. (CCF-A, JCR Q1)
- **Peichun Li**, Huanyu Dong, Liping Qian, Sheng Zhou, and Yuan Wu, "FlexGen: Efficient On-Demand Generative AI Service with Flexible Diffusion Model in Mobile Edge Networks," *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, no. 2, pp. 961-973, Apr. 2025. (JCR Q1)
- **Peichun Li**, Guoliang Cheng, Xumin Huang, Jiawen Kang, Rong Yu, Yuan Wu, and Miao Pan. "AnycostFL: Efficient On-Demand Federated Learning over Heterogeneous Edge Devices," in *Proceedings of IEEE INFOCOM*, 2023. (CCF-A Conference)
- **Peichun Li**, Guoliang Cheng, Xumin Huang, Jiawen Kang, Rong Yu, Yuan Wu, Miao Pan, and Dusit Niyato. "Snowball: Energy Efficient and Accurate Federated Learning with Coarse-to-Fine Compression over Heterogeneous Edge Devices," *IEEE Transactions on Wireless Communications*, vol. 22, no. 10, pp. 6778-6792, Oct. 2023. (JCR Q1)
- **Peichun Li**, Yupei Zhong, Chaorui Zhang, Yuan Wu, and Rong Yu. "FedRelay: Federated Relay Learning for 6G Mobile Edge Intelligence," *IEEE Transactions on Vehicular Technology*, vol. 72, no. 4, pp. 5125-5138, April 2023. (JCR Q1)
- **Peichun Li**, Xumin Huang, Miao Pan, and Rong Yu. "FedGreen: Federated Learning with Fine-Grained Gradient Compression for Green Mobile Edge Computing," in *Proceedings of IEEE Global Communications Conference (GlobeCom)*, 2021.
- **Peichun Li**, Guoliang Cheng, Jiawen Kang, Rong Yu, Liping Qian, Yuan Wu, and Dusit Niyato. "FAST: Fidelity-Adjustable Semantic Transmission over Heterogeneous Wireless Networks," in *Proceedings of IEEE ICC*, 2023.
- Rong Yu (Advisor) and **Peichun Li**. "Toward Resource-Efficient Federated Learning in Mobile Edge Computing," *IEEE Network*, vol. 35, no. 1, pp. 148-155, January/February 2021. (JCR Q1)

Academic Services

- Technical Program Committee (TPC) Member for leading IEEE conferences, including IEEE Globecom, HPCC, ICCC, INFOCOM ICCN, and VTC.
- Reviewer for journals such as IEEE TMC, IEEE TWC, IEEE Network, IEEE TVT, IEEE TNSE, IEEE TCCN, IEEE IoTJ, IEEE/CIC China Communications, and Elsevier Computer Networks, among others.